

Risk-Sensitive Decision Making in MDPs

Ruszczynski (2010) & Ruszczynski–Shapiro (2006)

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The Classical MDP Objective

Standard infinite-horizon discounted MDP:

$$\min_{\Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \alpha^{t-1} c(x_t, u_t) \right], \quad \alpha \in (0, 1)$$

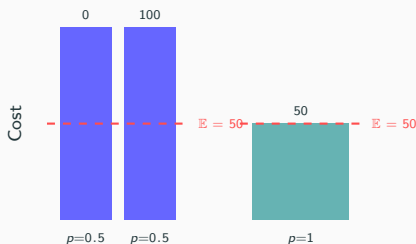
Solved by the Bellman equation:

$$v(x) = \min_{u \in U(x)} \{ c(x, u) + \alpha \mathbb{E}[v(x_{t+1}) \mid x_t = x, u_t = u] \}$$

Hidden assumption. Expected value treats two cost distributions as identical if they share the same mean, regardless of how spread out or tail-heavy they are: Risk-Neutrality

Why Expected Value Is Not Enough

A: Random $\{0,100\}$ **B:** Certain 50



Both scenarios: $\mathbb{E}[\text{cost}] = 50$.

Scenario A: $\text{cost} \in \{0, 100\}$ with prob. 0.5 each

Scenario B: certain cost of 50

The expected value model calls these *identical*.

A risk-averse agent — concert promoter, airline, device operator — is **not** indifferent.

Risk neutrality ignores the *shape* of the distribution. We need richer criteria.

Static Risk Measures

A static risk measure maps a Random Cost to a single risk-adjusted number. Consider two different portfolios with different loss profiles:

Portfolio A (Predictable Loss):

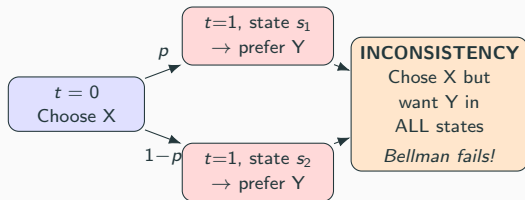
- 95% chance: \$0 loss.
- 5% chance: Exactly \$100 loss.

Portfolio B (Catastrophic Loss):

- 95% chance: \$0 loss.
- 4% chance: \$100 loss.
- 1% chance: \$1,000 loss (market crash).

Measure	Port. A	Port. B	Interpretation
Mean (Risk Neutral)	\$5	\$14	Only cares about the weighted average; doesn't distinguish between "stable" and "volatile."
VaR (95.01% conf.)	\$100	\$100	Blind to the extreme. It only sees that the "bad outcomes" start at \$100. It cannot see the \$1,000 disaster hidden in the 1% tail.
CVaR (95% conf.)	\$100	\$280	Sensitive to the extreme. It averages the worst 5%. It successfully identifies that Portfolio B is nearly 3x as dangerous as A.
Entropic Risk	\$8.24	\$540.00	Uses an exponential penalty $\rho(X) = \frac{1}{\theta} \log \mathbb{E}[e^{\theta X}]$, very sensitive to the extreme outlier in B.

Why Static Risk Measures Break in Dynamic Problems



Fix: require the nested structure (Thm. 1, Ruszczyński 2010)

- **Static use:** $\text{VaR}(c_1 + \dots + c_T)$ ignores information arrival $\rightarrow$ optimal plan today is not optimal later.
- **Naive dynamic use:** recomputing conditional VaR at each step changes action rankings $\rightarrow$ preferences shift over time.

Example: The problem with VaR

At $t = 0$: VaR prefers policy X .

At $t = 1$: in *every* state, VaR prefers Y .

The agent chose X , but at $t = 1$ they regret it and want to switch.

Consequence: Bellman's principle of optimality fails. We cannot decompose the problem backwards through time.

Time Consistency

If Z weakly preferred to W from all future vantage points, then preferred today:

$$\rho_{\theta, T}(Z_{\theta}, \dots) \leq \rho_{\theta, T}(W_{\theta}, \dots)$$

$$\Rightarrow \rho_{\tau, T}(Z) \leq \rho_{\tau, T}(W)$$

The Exception: Entropic Risk Measure

The **Entropic Risk Measure** is a unique static measure that is time consistent. Using an exponential disutility function $V(x) = e^{\theta x}$, the deterministic equivalent cost for a policy g is:

$$\bar{J}_\theta(g) = \frac{1}{\theta} \log \mathbb{E} \left[\exp \left(\theta \sum_{t=1}^T c(X_t, U_t) \right) \right]$$

Multiplicative DP Decomposition

Thanks to the properties of the exponential (converting additive costs over time into multiplicative factors), we get a valid multiplicative DP decomposition (see 31.1 in online notes):

$$V_t(s) = \min_{u \in \mathcal{U}} \left\{ \exp(\theta c(s, u)) \sum_{y \in \mathcal{S}} P_{sy}(u) V_{t+1}(y) \right\}$$

Terminal Condition: $V_{T+1}(s) = 1$ (representing $e^{\theta \cdot 0}$).

The Goal: We need to move to a **general theory** of risk measures for sequential problems, so we can handle dynamic risk without relying strictly on the unique mathematical quirks of the exponential function for example.

Axioms for One-Step Conditional Risk Mapping

Let $\mathcal{F}_t \subset \mathcal{F}_{t+1}$ be σ -algebras of subsets of a set Ω . Let $\mathcal{Z}_t \subset \mathcal{Z}_{t+1}$ be linear spaces of real-valued functions measurable with respect to $\mathcal{F}_t$ and $\mathcal{F}_{t+1}$, respectively. A mapping $\rho_t : \mathcal{Z}_{t+1} \rightarrow \mathcal{Z}_t$ is a **conditional risk mapping** if for future costs $Z, W \in \mathcal{Z}_{t+1}$ the following hold:

(A1) Convexity

$$\rho_t(\lambda Z + (1-\lambda)W) \leq \lambda \rho_t(Z) + (1-\lambda)\rho_t(W)$$

Diversification never increases risk ($\lambda \in [0, 1]$).

(A2) Monotonicity

$$Z \leq W \Rightarrow \rho_t(Z) \leq \rho_t(W)$$

Outcomes that are always less costly are assessed as less risky.

(A3) Predictable Translation Equivariance

$$\rho_t(V + W) = V + \rho_t(W), \quad \forall V \in \mathcal{Z}_t$$

A cost V known at time t shifts risk by exactly that amount—no interaction with future uncertainty.

(A4) Positive Homogeneity

$$\rho_t(\beta Z) = \beta \rho_t(Z), \quad \beta \geq 0$$

Scaling costs scales risk proportionally.

These are the axioms for a conditional risk measure, applied *conditionally* at each step: the risk mapping ρ_t takes an uncertain future cost from tomorrow ($\mathcal{Z}_{t+1}$) and translates it into a risk-adjusted value for today ($\mathcal{Z}_t$), conditioned on what is observed up to time t .

Theorem 1 — The Nested Structure

Condition (7) — Translation Equivariance

$$\rho_{t,T}(Z_t, Z_{t+1}, \dots) = Z_t + \rho_{t,T}(0, Z_{t+1}, \dots)$$

Known current cost shifts risk by that amount only; no interaction with future uncertainty.

Condition (8) — Normalisation

$$\rho_{t,T}(0, 0, \dots, 0) = 0$$

Zero costs carry zero risk. Together with (7): the measure evaluates only future uncertainty.

Theorem 1 (Ruszczynski 2010)

Under conditions (7)–(8), $\{\rho_{t,T}\}$ is time-consistent **iff** for all $\tau \leq \theta \leq T$:

$$\rho_{\tau,T}(Z_{\tau}, \dots, Z_T) = \rho_{\tau,\theta} \left(Z_{\tau}, \dots, \underbrace{\rho_{\theta,T}(Z_{\theta}, \dots, Z_T)}_{\text{pre-collapsed}} \right) \quad (9)$$

One-step factorisation (apply 9 repeatedly):

$$\rho_{1,T} = Z_1 + \rho_1(Z_2 + \rho_2(\dots + \rho_{T-1}(Z_T))) \quad (10)$$

Key insight. The full horizon measure collapses into a *right-to-left chain*: each ρ_t compresses the tail into a single number that feeds into ρ_{t-1} . This chain structure is what restores Bellman backward induction.

Two Canonical Examples

Mean-Semideviation (Eq. 11)

$$\rho_t(Z_{t+1}) = \mathbb{E}[Z_{t+1}|\mathcal{F}_t] + \kappa \left(\mathbb{E}[(Z_{t+1} - \mathbb{E}[Z|\mathcal{F}_t])_+^r | \mathcal{F}_t] \right)^{1/r}$$

- $\kappa = 0$: collapses to $\mathbb{E}$ (risk-neutral)
- $\kappa = 1$: maximum upside penalty
- Penalises outcomes *above* the mean only
- $r = 1$: linear; $r = 2$: semivariance

Conditional CVaR (Eq. 12)

$$\rho_t(Z_{t+1}) = \inf_{U \in \mathcal{Z}_t} \left\{ U + \frac{1}{\alpha} \mathbb{E}[(Z_{t+1} - U)_+ | \mathcal{F}_t] \right\}$$

- Cond. mean of worst α -fraction given $\mathcal{F}_t$
- $\alpha \rightarrow 1$: risk-neutral
- $\alpha \rightarrow 0$: worst-case (minimax)
- Solved as a small LP in practice

Both satisfy (A1)–(A4) and are time-consistent when composed via eq. (10). Neither requires modifying the MDP structure — only the one-step evaluation σ changes.

From Conditional to Markov Risk Measures

Problem. ρ_t is $\mathcal{F}_t$ -measurable and could depend on the full history $(x_1, \dots, x_t)$, destroying the Markov property and Bellman tractability.

Solution. Restrict ρ_t to depend only on x_t and $Q_t(x_t, u_t)$.

Definition 5 — Risk Transition Mapping

A measurable functional $\sigma : \mathcal{V} \times X \times \mathcal{M} \rightarrow \mathbb{R}$ such that

$$\sigma(v, x, Q(x, u)) = \sup_{\mu \in \mathcal{A}(x, Q(x, u))} \langle v, \mu \rangle$$

where $\langle v, \mu \rangle = \int_X v(y) \mu(dy)$ and $\mathcal{A}(x, Q(x, u)) \subset \mathcal{M}$ is the **risk envelope**.

Definition 6 — Markov Risk Measure

ρ_t is a **Markov risk measure** if there exists a risk transition mapping σ_t such that

$$\rho_t(v(x_{t+1})) = \sigma_t(v, x_t, Q_t(x_t, u_t))$$

The measure depends on current state and action only — the Markov property is restored.

Definition 6 is the bridge: takes an abstract coherent ρ_t satisfying (A1)–(A4) and gives it a *computable*, state-dependent form compatible with dynamic programming.

Risk Aversion as Kernel Ambiguity

Risk-neutral: $\mathcal{A} = \{Q(x, u)\}$ — singleton.

One kernel, trusted completely.

$$\sigma = \langle v, Q \rangle = \mathbb{E}[v(x_{t+1})]$$

$$\mathcal{A} = \{Q\}$$

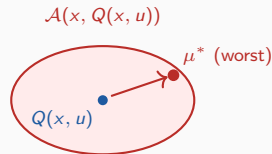


No adversary

Risk-averse: $\mathcal{A}$ is a set of kernels.

An adversary picks the worst μ^* from the envelope.

$$\sigma = \sup_{\mu \in \mathcal{A}(x, Q(x, u))} \langle v, \mu \rangle$$



Larger $\mathcal{A}$ = more pessimism = stronger risk aversion. When $\mathcal{A}$ shrinks to a point, we recover the risk-neutral expected value.

The Risk-Averse Bellman Equation

One substitution. Replace $\mathbb{E}[v(x_{t+1}) \mid x_t, u_t]$ with $\sigma(v, x_t, Q_t(x_t, u_t))$.

Theorem 2 — Finite Horizon

$$v_{T+1}(x) = c_{T+1}(x)$$

$$v_t(x) = \min_{u \in U_t(x)} \left\{ c_t(x, u) + \underbrace{\sigma_t(v_{t+1}, x, Q_t(x, u))}_{\text{replaces } \mathbb{E}[v_{t+1}(x_{t+1})]} \right\}$$

for $t = T, \dots, 1$.

An optimal Markov policy $\hat{\Pi} = \{\hat{\pi}_1, \dots, \hat{\pi}_T\}$ exists with:

$$\hat{\pi}_t(x) \in \operatorname{argmin}_{u \in U_t(x)} \{c_t(x, u) + \sigma_t(v_{t+1}, x, Q_t(x, u))\}$$

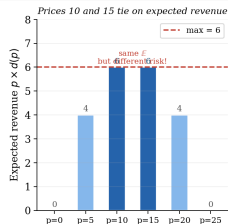
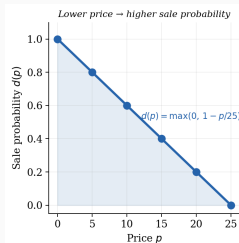
Conditions (i)–(v)

- (i) $Q_t(x, \cdot)$ continuous for every $x \in X$, $t = 1, \dots, T$
- (ii) ρ_t Markov; multifunctions $\mathcal{A}_t(x, \cdot)$ lower semicontinuous for every $x \in X$
- (iii) $x \mapsto c_t(x, u_t(x))$ and $c_{T+1}(\cdot)$ lie in $\mathcal{V}$ for all measurable selections u_t
- (iv) $c_t(x, \cdot)$ lower semicontinuous for every $x \in X$
- (v) $U_t(x)$ compact for every $x \in X$

Summary. Same algorithm as classical DP. Replace $\mathbb{E}[\cdot]$ with $\sigma(\cdot)$. The choice of σ (CVaR, MSD, minimax, ...) determines the agent's pessimism level — and changes the optimal policy.

Risk-Averse Dynamic Pricing: The Concert Problem

Setup. A firm has $C = 5$ tickets and $T = 10$ periods to sell them. At each period it sets price $p \in \{0, 5, 10, 15, 20, 25\}$. Demand: $d(p) = \max(0, 1 - p/25)$. Unsold tickets at period 10 are worth **zero**.

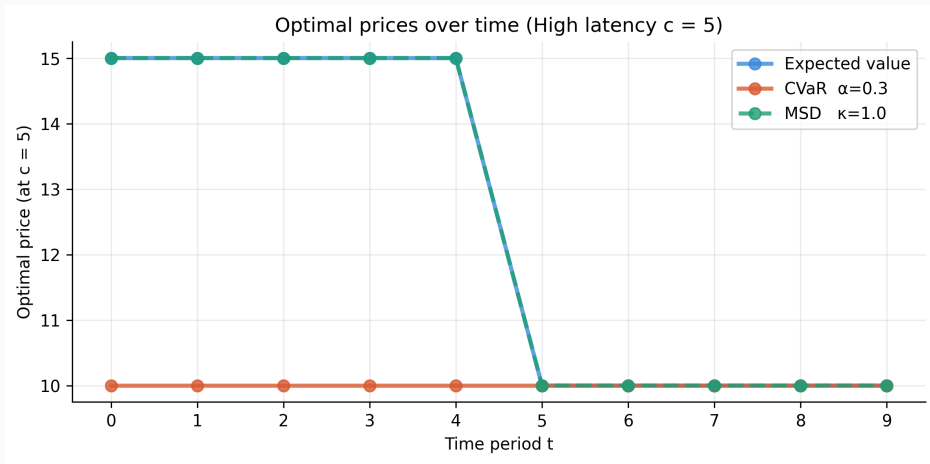


The core tension.

p	$d(p)$	$p \times d(p)$
0	1.0	0
5	0.8	4
10	0.6	6
15	0.4	6
20	0.2	4
25	0.0	0

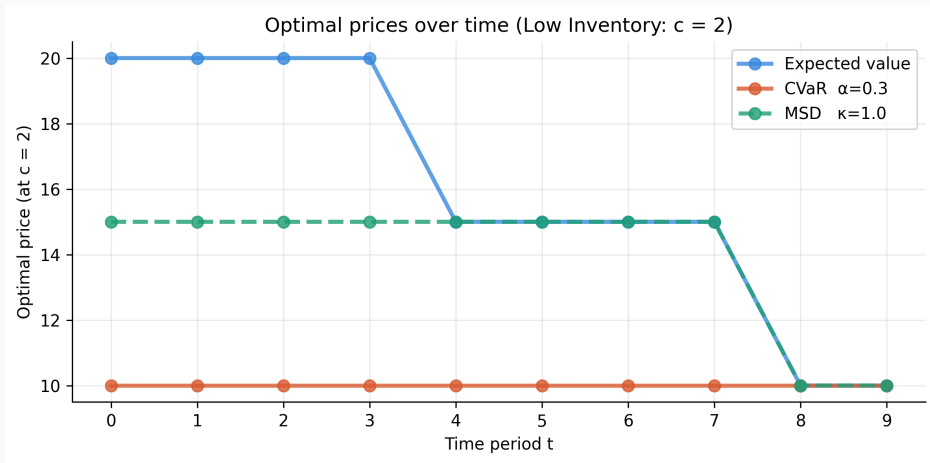
Prices 10 and 15 **tie** on single-period expected revenue. But price 15 earns nothing 60% of the time; price 10 earns nothing only 40% of the time. **Same mean, very different tail risk.**

Pricing Policy — Starting Inventory $c = 5$



CVaR sets **lower prices** throughout — raising sale probability to avoid ending with unsold inventory.

Pricing Policy — Starting Inventory $c = 2$



With fewer units, risk aversion is even more pronounced — CVaR keeps prices low from the outset.

The One-Line Summary

Replace $\mathbb{E}[v(x_{t+1}) \mid x_t, u_t]$ with

$$\sigma(v, x_t, Q(x_t, u_t)) = \sup_{\mu \in \mathcal{A}(x_t, Q_t)} \langle v, \mu \rangle$$

Same algorithm. Different σ . Different policy.

Markov Structure Preserved

Even under risk aversion, an optimal policy that is **Markov** always exists (Thm. 2, Ruszczynski 2010). The agent needs to track only the current state x_t — not the full history.

Control Kernel $\rightarrow$ Risk Envelope

The single trusted kernel $Q(x, u)$ of classical DP is replaced by a set of kernels $\mathcal{A}(x, Q(x, u))$. An adversary selects the worst $\mu^* \in \mathcal{A}$ — the size of $\mathcal{A}$ controls the degree of risk aversion.

Key Results — Ruszczynski (2010)

Thm. 1 Nested structure $\Leftrightarrow$ time consistency

Def. 5–6 Risk transition mapping σ restores the Markov property

Thm. 2 Risk-averse Bellman equations; optimal Markov policy $\hat{\Pi}$ exists

- Ruszczyński (2010) *Mathematical Programming* 125(2): 235–261
- Ruszczyński & Shapiro (2006) *Math. Oper. Res.* 31(3): 544–561
- Schlosser & Gönsch (2023) *EJOR* 310: 1151–1163